

B. Tech Degree VI Semester Examination, April 2009

ME 603 MACHINE DESIGN I (2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Use of approved design data book permitted)
(Assume reasonable data wherever necessary)

PART - A (Answer ALL questions)

(8 x 5 = 40)

- I. (a) Explain the term 'stress concentration'. How it affects the design? What are the solutions to avoid stress concentration?
- (b) Define the term 'creep'. Explain its mechanism. Give an application where creep is a design factor.
- (c) What is meant by self locking property of threads? Obtain the expression for efficiency of self-locking screws.
- (d) Two shafts are connected by means of a coupling to transmit 5 HP at 1440 rpm. The flanges of the coupling are fastened by means of 4 bolts at a radius of 30 mm. Permissible shear stress in bolts = 30 N/mm^2 . Design the bolts.
- (e) With neat sketch, explain the term 'Rivet'. Also explain the different failure modes in riveted joints.
- (f) Give the classification of springs. Explain the term surge in springs. How is it eliminated?
- (g) Sketch the different types of welded joints. Give the advantages and disadvantages of welded joints.
- (h) Explain the term 'critical speed of shaft'. How it affects the life of shaft?

PART - B

(4 x 15 = 60)

- II. (a) Determine maximum stress induced in the following cases taking stress concentration into account :
 - (i) A rectangular plate with a hole under an axial load of 10 kN
 - (ii) The circular shaft with a step under an axial load of 10 kN
 - (iii) The shaft under a bending moment of 50 Nm
 - (iv) The shaft under a twisting moment of 50 Nm.
- (b) In an elastic material, principal stresses are tensile and compressive and the ratio being 4:1. Determine the limiting stress according to different theories of failure if tension test gives elastic limit of the material as 400 N/mm^2 . Assume poisson's ratio as 0.3.

OR

- III. (a) Determine maximum stress induced in the shaft when transmitting 20 KW at 400 rpm. The diameter of the shaft is 40 mm. Take stress concentration into account.



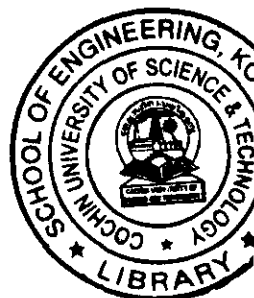
- (b) A simple tension test and rotating beam test on a ductile material gave the following results :

Material C 40 steel annealed

Diameter of specimen = 12.5 mm, Yield strength = 240 N/mm^2

Ultimate strength = 450 N/mm^2 , Endurance limit = 180 N/mm^2

Determine the design stress for the following working conditions :



(Turn Over)

- (i) For trusses where the loading is almost constant. The size of the member is nearly 40 mm. Factor of safety is 2
- (ii) For a piston rod where the load is completely reversed. The surface of the rod is ground, no stress concentration and factor of safety is 2
- (iii) For bolts in a cylinder cover where the load is fluctuating due to gas pressure. Maximum load on the bolt is 5t and minimum load is 3t. Since load is unpredictable, FOS is 3. The surface of bolt is hot rolled. (10)

- IV. Design a sleeve and cotter joint to connect two rods which transmits maximum tensile load of 100 kN. Assume that the rods, sleeve and cotters are made of same material and the permissible stresses in the material are 65 N/mm² (in tension), 130 N/mm² (in compression) and 50 N/mm² (in shear) make a neat dimensioned sketch of the joint. (15)

OR

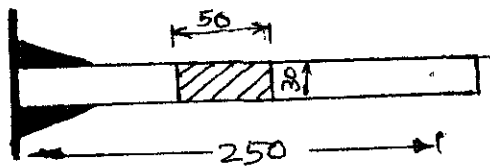
- V. Design a knuckle joint to withstand a tensile load of 60 kN using steel with the following permissible stresses :
 $\sigma_t = 62.5 \text{ N/mm}^2, \sigma_c = 75 \text{ N/mm}^2, \tau = 50 \text{ N/mm}^2$
 Make a neat sketch of the joint. (15)

- VI. Design a riveted joint for the longitudinal and circumferential seam of a boiler 2 meters diameter to withstand maximum pressure of 100 N/cm² (gauge). The ultimate strength of plate 370 N/mm² in tension and a factor of safety of 5 is recommended. The allowable stress in shearing of rivets is 60 N/mm² and in crushing of rivets is 100 N/mm². Sketch the joint. (15)

OR

- VII. A semi elliptical spring 1500 mm long carries a total load of 12 kN. The spring is to have 10 leaves, two being full length leaves. The deflection is to be 60 mm. Design and sketch the spring. (15)

- VIII. Determine the size of the weld for a bar when it is loaded in the following bar :
 (i) Subjected to axial load of 75 kN
 (ii) Subjected to bending moment of 500 N – mm. Allowable stress in weld is 90 N/mm².
 (iii) Determine stress in the weld in case (ii) if the same bending moment is produced by a load of 2000 N at the end. (15)



OR

- IX. A solid shaft is supported on two bearings 1800 mm apart and rotates at 250 rpm. A 20° involute gear D, 300 mm diameter is keyed to the shaft at a distance of 150 mm to the left of the right hand bearing. Two pulleys B and C are located on the shaft at distance 600 mm and 1350 mm respectively to the right of the left hand bearing. The diameters of the pulleys B and C are 750 mm and 600 mm respectively. 40 HP is supplied to the gear out of which 25 HP is taken of in the pulley C and 15 HP from B. The drive from B is vertically downward while from C, the drive is downward at an angle of 60° to the horizontal. In both cases the tension ratio is 2 and the angle of lap is 100°. The combined fatigue and shock factors for torsion and bending may be taken as 1.5 and 2.0 respectively. Design a suitable shaft taking working stress to be 42 N/mm² in shear at 84 N/mm² in tension. (15)
